Chapter 26

Particle Properties of Light: Photon

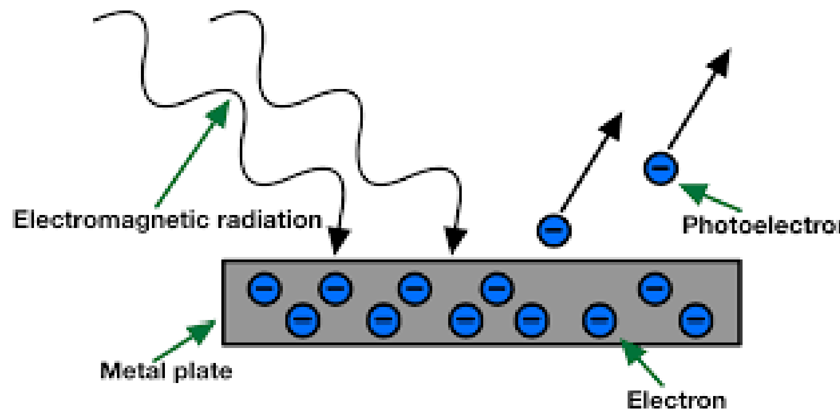
In this chapter we will study the following:

- **Photoelectric effect and particle nature of light**
- **Photons**
- **X-ray**
- **Compton effect**
- **Wave-particle Duality**

Chapter 26

Particle Properties of Light: Photon

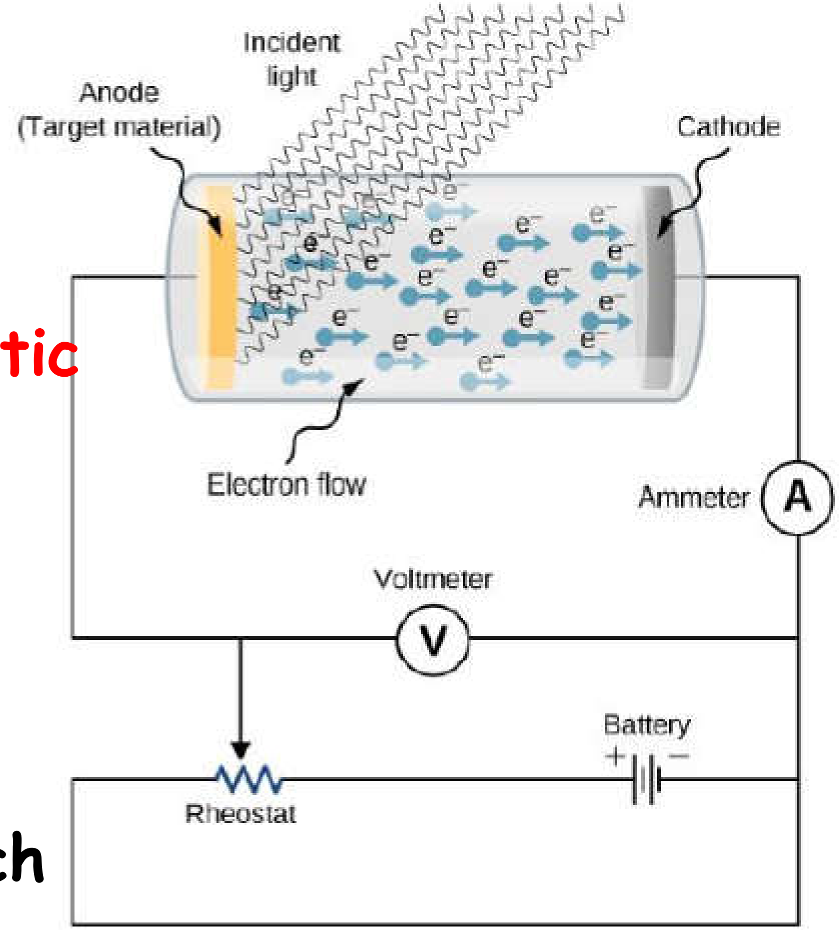
26.1 Photoelectric Effect:



When light shines on a metal, electrons can be ejected from the surface of the metal in a phenomenon known as the photoelectric effect. This process is also often referred to as photoemission, and the electrons that are ejected from the metal are called photoelectron

Experimental setup:

The target material serves as the anode, which becomes the emitter of photoelectrons when it is illuminated by monochromatic radiation



photocurrent in a circuit, current that flows when a photoelectrode is illuminated

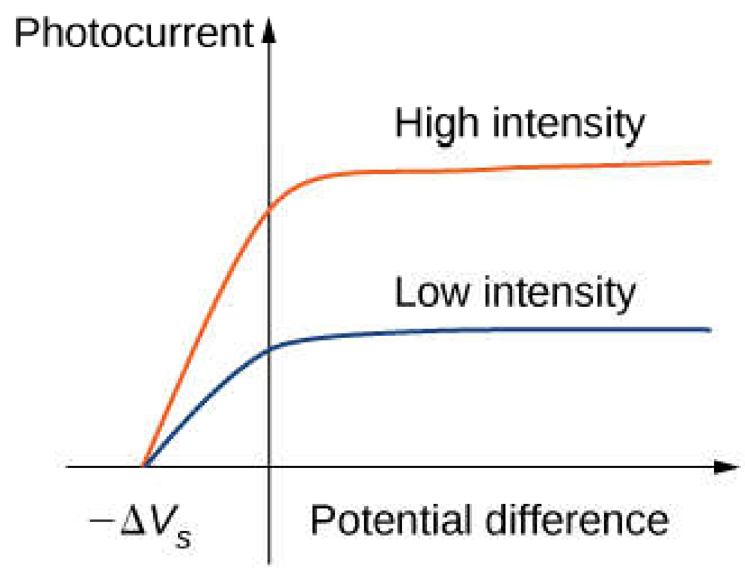
The potential difference at which the photocurrent stops flowing is called the stopping potential

Light behaves as a particle

The largest (maximum) kinetic energy of photoelectrons can be directly measured by measuring the stopping potential

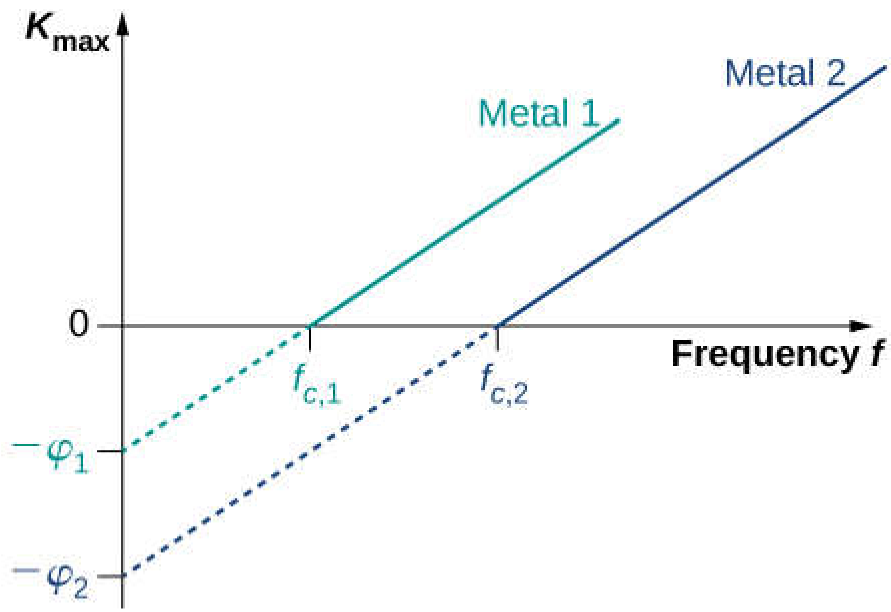
$$K_{max} = e\Delta V_s .$$

The experiment shows that the maximum kinetic energy of photoelectrons is independent of the light intensity.



The presence of a cut-off frequency (Threshold Frequency f_o)

For any metal surface, there is a **minimum frequency** of incident radiation below which photocurrent does not occur.
Cut-off frequency of incident light below which the photoelectric effe



A side is the Kinetic energy of photoelectrons at the surface versus the frequency of incident radiation

$$K_{max} = hf - \phi$$

See next slide

The Work Function: ϕ

The photoelectric effect was explained in 1905 by A. Einstein

energy of a photon $$E_f = hf$$

h = Planck's constant = 6.63×10^{-34} Js
f = frequency of the light

The energy needed to detach photoelectron from the metal surface is called the Work function ϕ

The question relate the kinetic energy to ϕ is given by

$$K_{max} = hf - \phi$$

Threshold Frequency: The minimum frequency of light required to liberate electrons from a material is determined by the material's binding energy. If the frequency of the incoming light is **below** this threshold frequency, the photons lack the necessary energy to release electrons from the material's surface, regardless of the intensity of the light. However, if the frequency of the incoming light is **equal to or greater** than the threshold frequency f_o, electrons can be emitted from the material. f_o = threshold frequency is given by

$$f_o = \frac{\varphi}{h}$$

So, the previous equation becomes:

$$K_{max} = hf - \phi = hf - hf_o$$

NOTE: Check questions 26.1, 26.2, 26.5, 26.14, 26.17 set#(8)

EXAMPLE: Radiation with a frequency of 3.13 $\times 10^{16}$ Hz releases an electron from a copper plate. The kinetic energy of the electron is 2 $\times 10^{-17}$J. Calculate the work function of the plate. Express your answer in ev.

Solution:

$$K_{max} = hf - \phi$$

$$\phi = hf - K_{max}$$

$$\phi = 6.63\times10^{-34}\times3.12\times10^{16} - 2\times10^{-17}$$

$$=7.52\times10^{-19}J$$
$$=7.52\times10^{-19}J\times(1ev/1.6\times10^{-19}J)$$
$$=4.7ev$$

Photons:

Quantum theory describes light as a particle called photon with wavelength λ.

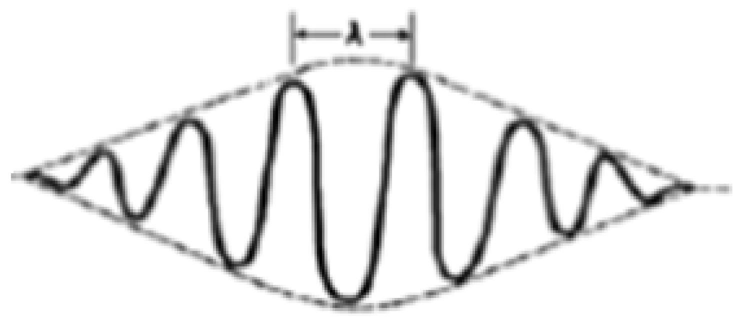
The energy of the photon is given by:

$$E = hf = hc/\lambda$$

Where h is planck constant = 6.63×10^{-34}Js

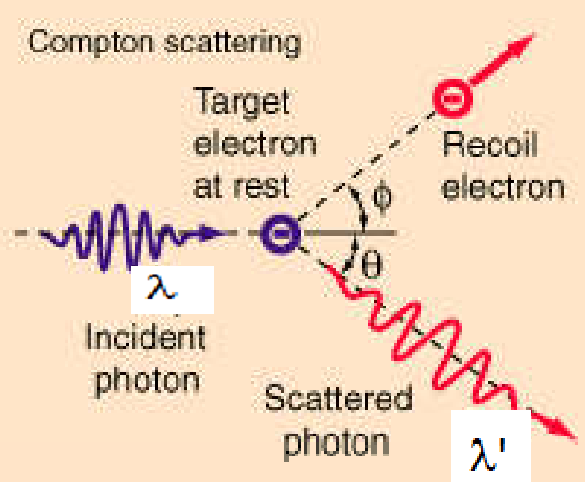
, c is the speed of light = 3×10^{8}m/s

and f is the frequency of the light



Representation of a Photon

26.3 X-Rays: Compton Scattering



Compton Scattering

Arthur H. Compton observed the scattering of x-rays from electrons in a carbon target and found scattered x-rays with a longer wavelength than those incident upon the target. The shift of the wavelength increased with scattering angle according to the Compton formula:

The scattered photon has longer wavelength than the incident photon means that there is a reduction depended on the scattered angle. This change in the wavelength is called Compton shift

In this process Light behaves as a particle

Compton effect was first observed by Arthur Compton in 1923 and this discovery led to his award of the 1927 Nobel Prize in Physics. The discovery is important because it demonstrates that light cannot be explained purely as a wave phenomenon. Compton's work convinced the scientific community that light can behave as a stream of particles (photons) whose energy is proportional to the frequency.

The change in wavelength of the scattered photon is given by:

$$hf' = hf - E_{el}$$

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - Cos\,\theta)$$

f' is the scattered frequency
f is the incident frequency
E_{el} is the energy of recoil electron

Compton wavelength

$$\lambda_C^e = \frac{h}{m_e c} \approx 2.43 \times 10^{-12} m$$

λ' is the scattered frequency
λ is the incident frequency
m_e is the electron mass
C is the speed of light

EXAMPLE: In the Compton scattering experiment the Incident X-ray have frequency of 10^{20}Hz at certain angle The outgoing X-ray have frequency 8×10^{19}Hz. Find the Energy of the recoiling electron in electron volts.

Solution: $E_{el} = hf - hf'$

$=h(f-f')$

$=6.63\times10^{-34}(10^{20} - 8\times10^{19})$

$=1.326\times10^{-14}J$

$= 1.326\times10^{-14}J\times(1\ ev/1.6\times10^{-19}J)$

$=82875ev$

EXAMPLE: Determine the change in the photon's wavelength That occurs when an electron scatters and X-ray photon (a) At θ =180° and (b) θ = 30°

Solution:

$$\Delta\lambda=\lambda_c(1-\cos\theta)$$

(a) $\Delta\lambda=2.43\times10^{-12}\ \text{m}\ (1-\underbrace{\cos180^\circ}_{\boxed{-1}})$

$$\Delta\lambda=4.86\times10^{-12}\ \text{m}$$

(b) $\Delta\lambda=2.43\times10^{-12}\ \text{m}\ \underbrace{(1-\cos30)}_{\boxed{0.134}}$

$$\Delta\lambda=(0.134)\times(2.43\times10^{-12}\ \text{m})$$

$$\Delta\lambda=3.26\times10^{-13}\ \text{m}$$

EXAMPLE: Calculate the energy of a photon with a frequency of 4 x10^{14}Hz.

Solution:

$$E = hf$$

Substituting in the values we get:

$$E = 6.63 \times 10^{-34} \times 4 \times 10^{14}$$
$$E = 2.652 \times 10^{-19}$$
$$E = 2.65 \times 10^{-19} J$$